

Module 1

Robertson, Rashai

Bellevue University, CSD-380

Assignment 1.3

June 7, 2026

DevOps stands for Development and Operations. It is a modern software development approach that combines development, operations, automation, collaboration, and continuous improvement. While DevOps became popular in the late 2000s, its history comes from several earlier movements, especially Lean, Agile, and Continuous Delivery (Vorel, 2025). Together, these movements changed how organizations build, test, release, and maintain software.

According to The Lean Enterprise Institute, the Lean movement, as we know it began through the Toyota Production System, which fed off Henry Ford's Flow Production. Toyota focused on eliminating waste, improving flow, reducing delays, and continuously improving processes. Lean taught organizations that value should be viewed from the customer's perspective. Anything that does not help deliver value to the customer should be reduced or removed.

This idea strongly influenced DevOps. In software development, waste can include waiting for approvals, fixing repeated defects, manual deployments, poor communication, and large unfinished projects. DevOps applies to Lean thinking by encouraging smaller batches of work, faster feedback, automation, and continuous improvement. Instead of waiting months to release software, DevOps teams try to move work smoothly from planning to development, testing, deployment, and operations.

Another major influence on DevOps was the Agile Manifesto, created in 2001 by a group of software developers in Snowbird, Utah (Project Management Institute, n.d.). Agile was a response to older, slower software methods that required long planning periods and heavy documentation before working software was delivered.

The Agile Manifesto values individuals and interactions, working software, customer collaboration, and responding to change. These values helped software teams become more flexible and customer focused. Agile encouraged teams to deliver software in small increments, receive feedback, and improve through repeated cycles.

However, Agile mainly improved the development side of software. Many organizations became faster at writing code, but they still struggled to release that code into production. Developers could finish work quickly, but operations teams often had to manage risky deployments, unstable systems, and last-minute problems. This gap between development and operations became one of the main problems DevOps attempted to solve.

The Continuous Delivery movement helped connect Agile development with reliable software releases. Continuous Delivery focuses on keeping software in a state where it can be safely released at almost any time (IBM, 2021). This requires automated testing, version control, deployment pipelines, and close collaboration between teams. The movement showed that releases should not be rare, stressful events. Instead, software delivery should become a normal, repeatable, and low-risk process. Continuous Delivery became one of the technical foundations of DevOps. It gave organizations practical methods for automating builds, tests, and deployments. This allowed teams to release changes more often while reducing errors.

The history of DevOps shows that it did not appear from nowhere. It developed from earlier movements that focused on better ways of working. Lean contributed to the ideas of eliminating waste, improving flow, and continuous improvement. Agile contributed flexibility, collaboration, and fast feedback. Continuous Delivery contributed automation and reliable release practices.

Together, these ideas shaped DevOps into a major approach for modern technology organizations. DevOps helps teams move away from slow, separated, and risky processes. Instead, it promotes collaboration, automation, shared responsibility, and faster delivery of value to customers.

References:

IBM. (n.d.). *What is continuous delivery?* IBM. <https://www.ibm.com/topics/continuous-delivery>

Lean Enterprise Institute. (n.d.). *A brief history of lean*. Lean Enterprise Institute. <https://www.lean.org/explore-lean/a-brief-history-of-lean/>

Project Management Institute. (n.d.). *The Agile Manifesto*. Project Management Institute. <https://www.pmi.org/disciplined-agile/agile/theagilemanifesto>

Vorel, R. (2025). The Evolution of DevOps. In: NoOps. Apress, Berkeley, CA. https://doi.org/10.1007/979-8-8688-1694-9_1